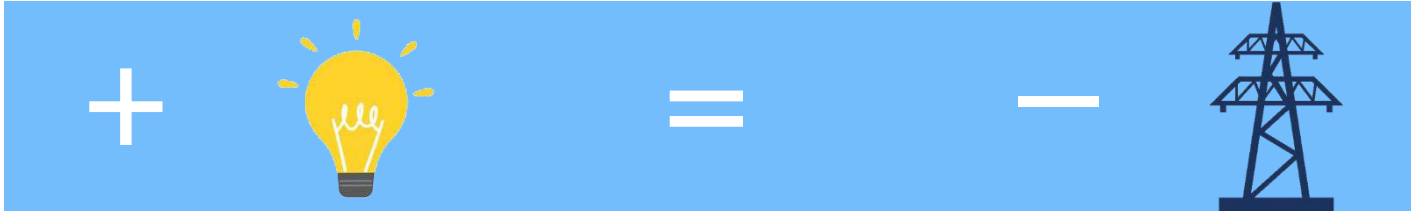


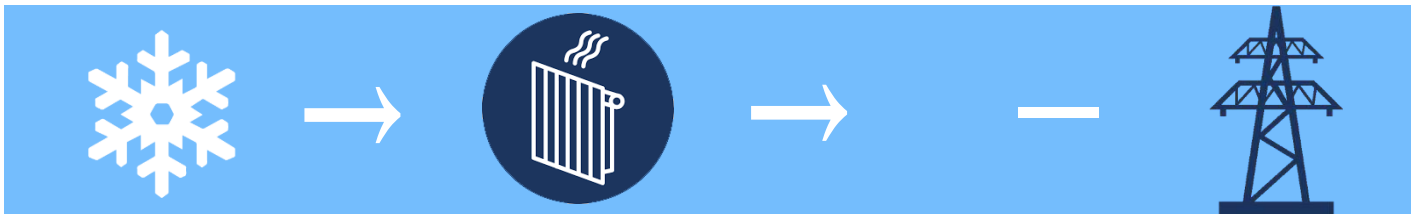
Temperature and forecast deviations

GENERAL INFO | The underlying time series (consumption deviation, see file) represents the physical deficit or the surplus of the electricity consumption in France.

EXPLANATION | More consumption than expected = deficit = negative value;
less consumption than expected = surplus = positive value



THESIS | The assumption is that the average daily consumption deviation value on one day can be explained by the temperature deviation in France on that day where, depending on the temperature deviation, the daily consumption deviation is on the average negative (deficit) or positive (surplus). Why? Utilities are doing their daily consumption forecast based on the historical consumption for the same day. Here's an example: If they do a forecast for the consumption for 01.01.2017, they will look at the consumption of 01.01. of 2016, 01.01. of 2015, 01.01. of 2014... etc. That is why we expect for example a deficit in the electricity grid in a cold winter (because of more heating) and a surplus in a warm winter. We expect the reverse relationship in summer.



QUESTION | Can we say with a certain **probability** that we will have a surplus or a deficit in the grid depending on the temperature deviation?

TASK | For this, please calculate the deviation between the daily temperatures for 2017 and the average temperature of these days in the past - you can choose the number of years you take to calculate the average. Compare the temperature deviations to the average daily consumption deviations.

ADVICE | **Make a table for summer and winter with several temperature deviation intervals.**

SOURCE | You will find temperatures of metering stations in France on the following website:
https://mesonet.agron.iastate.edu/request/download.phtml?network=FR_ASOS.

You are free to choose between the weather stations. If you are facing problems gathering data, uncheck everything in "Limit Report Types".

Thank you for approaching this task.
If you have any questions, feel free to contact us.